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| <b>(a) Reproduction</b>                                                |                                                                                                                                                            |
| <b>Students should:</b>                                                |                                                                                                                                                            |
| 3.1                                                                    | understand the differences between sexual and asexual reproduction                                                                                         |
| 3.2                                                                    | understand that fertilisation involves the fusion of a male and female gamete to produce a zygote that undergoes cell division and develops into an embryo |
| <b>Flowering plants</b>                                                |                                                                                                                                                            |
| 3.3                                                                    | describe the structures of an insect-pollinated and a wind-pollinated flower and explain how each is adapted for pollination                               |
| 3.4                                                                    | understand that the growth of the pollen tube followed by fertilisation leads to seed and fruit formation                                                  |
| 3.5                                                                    | <i>practical: investigate the conditions needed for seed germination</i>                                                                                   |
| 3.6                                                                    | understand how germinating seeds utilise food reserves until the seedling can carry out photosynthesis                                                     |
| 3.7                                                                    | understand that plants can reproduce asexually by natural methods (illustrated by runners) and by artificial methods (illustrated by cuttings)             |
| <b>Humans</b>                                                          |                                                                                                                                                            |
| 3.8                                                                    | understand how the structure of the male and female reproductive systems are adapted for their functions                                                   |
| 3.9                                                                    | understand the roles of oestrogen and progesterone in the menstrual cycle                                                                                  |
| <b>3.10B understand the roles of FSH and LH in the menstrual cycle</b> |                                                                                                                                                            |
| 3.11                                                                   | describe the role of the placenta in the nutrition of the developing embryo                                                                                |
| 3.12                                                                   | understand how the developing embryo is protected by amniotic fluid                                                                                        |
| 3.13                                                                   | understand the roles of oestrogen and testosterone in the development of secondary sexual characteristics                                                  |